

Alexander Belik

Software Engineering | AI Systems | Scientific Computing

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PROFILE

Theoretical physics graduate and current M.Sc. High-Performance Computing student at Trinity College Dublin. I build research platforms, GPU and HPC simulation tooling, machine-learning systems, and full-stack products across theoretical physics, scientific computing, and data-heavy software.

EDUCATION

M.Sc. High-Performance Computing

Trinity College Dublin, current

B.A. Theoretical Physics

Trinity College Dublin, 2022-2026

EXPERIENCE

Scientific Software Developer

Trinity College Dublin

May 2026 - Sep 2026

- Built a digital lab notebook platform storing experimental measurements as structured, auditable records.
- Developed GPU-accelerated physics demonstrations that run headlessly on the Leonardo EuroHPC system.

Teaching Assistant

Trinity College Dublin

Oct 2025 - May 2026

- Led weekly tutorials and problem-solving sessions aligned with lectures.
- Graded assignments and exams with clear, constructive feedback.

Research Assistant

Trinity College Dublin, CRANN Institute

Jun 2025 - Sep 2025

- Designed Echo State Network reservoir models for quantum time-evolution data.
- Analysed model stability, convergence, and predictive accuracy across hyperparameters.

Software Engineer (EdTech)

Grinds360

Jun 2024 - Dec 2025

- Built Python automation tools for video and interactive lesson workflows.
- Contributed to UX iteration in Figma and full-stack product features.

Experimental / Computational Research Assistant

Trinity College Dublin

May 2024 - Sep 2024

- Created a Python spectroscopy control system for ferrofluids and ferronematic liquid crystals.
- Built automated acquisition and analysis workflows for repeated experimental data.
- Contributed to liquid-crystal research that led to co-authored publication work.

Exam Corrector

The Dublin Academy of Education

Jul 2021 - Jan 2024

- Corrected exams and coursework and supported marking workflows.

Selected work

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HOME PLATFORM

Inhabis (in progress)

A local-first operating layer for a connected home: one trusted model of the house, explicit policy, and evidence that an action actually worked.

- TypeScript server with separated layers for operational state, derived planning context, and constrained planning
- Embedded Matter controller plus an Android companion that performs BLE and Thread onboarding away from the host

Tools: TypeScript, Node.js, React, Python, Matter / Thread, Kotlin

DESKTOP WORKSPACE

Orion Mini (in progress)

Local-first Windows workspace for projects, tasks, reminders, transcription, and an assistant, with React rendering inside a Qt shell.

- Multi-track transcription library with checkpoints, corrections, anchored notes, and timestamped exports
- Activity-aware reminders with idle detection and escalating native alerts that cannot be quietly dismissed

Tools: Python, PySide6 / Qt, React, TypeScript, CUDA, Whisper

RESEARCH DATA PLATFORM

LabFlow

Digital lab notebook for a materials research institute that stores measurements as structured relational records.

- React and TypeScript frontend over a FastAPI and PostgreSQL backend with versioned migrations
- Sample, project, instrument, and booking models with role-based access across organisations and sites

Tools: React, TypeScript, FastAPI, PostgreSQL, Docker, Tailwind CSS

HPC AND GPU SIMULATION

Leonardo Visual Demos

Thirteen GPU-accelerated physics demonstrations that run headlessly on the Leonardo supercomputer and replay through a browser viewer.

- Thirteen solvers spanning lattice-Boltzmann flow, particle-mesh cosmology, N-body galaxy collisions, reaction-diffusion, plasma control, and molecular dynamics
- One demo contract that runs on NumPy, CuPy/CUDA, or a hybrid pipeline overlapping GPU solving with CPU frame encoding

Tools: Python, CuPy / CUDA, PyTorch, NumPy, SLURM, JavaScript

[View on GitHub](#)

LOCAL AI ASSISTANT

ORION (in progress)

Windows-first local assistant with wake-word gating, local LLM orchestration, memory, guarded tools, and a PyQt overlay/dashboard.

- LLM-first capability selection with typed tool boundaries
- Wake-word and follow-up metadata for safer utterance handling

Tools: Python, PyQt, Ollama, SQLite, Whisper, Local LLMs